

Data Flow Diagram

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	2 Marks

DFD Symbol Legend

Symbol	Name	Description
Green oval	External Entity	A source or sink outside the system: the Student (provides input, receives results) and the LLM Provider (Gemini / OpenRouter).
Blue rectangle, numbered	Process	An activity that transforms incoming data into outgoing data (e.g., Run BiLSTM Predictor, Decision Engine).
Yellow rectangle (D1/D2/D3)	Data Store	A place data is held for later use (Session State, Model Weights, Interaction Log CSV).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what is being passed.

Level-1 Data Flow Diagram

Traces one Analyze click end-to-end: from the student's field + problem text, through both models, the decision engine, AI response generation, and back to the student via the results panel and analytics dashboard.

